

Abdelrahman Tarek

Data Scientist

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Professional Summary

Electronics and Communications student with strong foundations in Machine Learning, Deep Learning, Computer Vision, and Generative AI. Skilled in building predictive models and data-driven dashboards using Python and Power BI. Currently advancing through intensive training program in Generative AI (DEPI). Passionate about applying AI to real-world problems in healthcare, business intelligence, and intelligent systems.

Professional Training & Programs

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| Generative AI Professional – Digital Egypt Pioneers Initiative (DEPI) | Dec 2025 – Present |
| <ul style="list-style-type: none">Studying core concepts of Generative AI, including LLMs, GANs, and diffusion-based models.Strengthened mathematical foundations (Statistics & Linear Algebra) for AI modeling.Built end-to-end ML & DL pipelines using Python for data science and AI applications.Worked with NLP models, attention mechanisms, Hugging Face, and transfer learning. | |
| Deep Learning for Computer Vision – DEY - Upskilling Programs (NTI) | Jan 2026 – Feb 2026 |
| <ul style="list-style-type: none">Implemented ANNs, CNNs, and modern deep learning architectures.Designed and trained image classification models using deep neural networks.Applied transfer learning with modern CNNs (ResNet, DenseNet).Performed image processing and feature extraction techniques. | |
| Machine Learning Intern – National Telecommunication Institute (NTI) | Jul 2025 – Aug 2025 |
| <ul style="list-style-type: none">Built ML pipelines for classification and regression problems.Applied feature engineering and evaluation metrics to optimize performance.Developed Python-based analytical tools for network and customer datasets. | |
| Machine Learning Intern – Elevvo Pathways | May 2025 – Jun 2025 |
| <ul style="list-style-type: none">Implemented supervised learning models using Scikit-learn.Automated data preprocessing for clinical datasets.Delivered a complete ML project on health risk prediction. | |
| Data Analyst Intern – DEY, National Telecommunication Institute (NTI) | May 2025 – Aug 2025 |
| <ul style="list-style-type: none">Cleaned, transformed, and visualized data using Python and Power BI.Queried large datasets using SQL for analytics and reporting.Designed executive-level dashboards. | |
| AI Career Essentials Intern – ALX Africa | Apr 2025 – Jun 2025 |
| <ul style="list-style-type: none">Applied generative AI tools for problem-solving and productivity.Studied ethical AI, deep learning fundamentals, and prompt engineering. | |

Education

Bachelor’s Degree in Electronics and Communications Engineering

Helwan University | Sep 2021 – Jul 2026
Accumulative Grade: Very Good

Technical Skills

- **Programming & Frameworks:** Python, NumPy, Pandas, Scikit-learn, TensorFlow, Keras
- **Data Analysis & Visualization:** Power BI, DAX, Power Query, SQL, Microsoft Excel, Data Cleaning & Wrangling
- **Machine Learning & AI:** Supervised & Unsupervised Learning, Deep Learning, CNNs, NLP, LLMs, Generative AI, Transfer Learning
- **Computer Vision:** Image Classification, Object Detection, Segmentation, Face Recognition

Projects

Graduation Project – NFC-Based Healthcare System

Tools: Python, FastAPI, MongoDB, NLP, REST APIs

- Developed a Drug–Drug Interaction (DDI) detection model using NLP to identify and analyze medication risks.
- Built a bilingual (Arabic/English) medical healthcare assistant chatbot for real-time patient support.
- Implemented backend services with FastAPI, and MongoDB to manage data and system integration.

EPL Match Result Prediction (Final Project – NTI Internship)

GitHub

Tools: Python, Pandas, Scikit-learn, XGBoost, Seaborn, Matplotlib

- Built an ML pipeline to predict match outcomes (Home / Draw / Away).
- Processed historical EPL data (2000–2024).
- Achieved best performance using **XGBoost**.

Adventure Works Sales Dashboard – Power BI

GitHub

Tools: Power BI, DAX, Power Query

- Cleaned sales data using advanced Power Query transformations.
- Developed KPIs for total sales, tax, subtotal, and order trends.
- Built interactive dashboards with KPIs, slicers, and drill-through.

Courses & Certifications

- **Future AWS AI Scientist** | Udacity Nanodegree Program
- **NVIDIA DLI Summer Training Program Generative AI** | ITI
- **Data Science Essentials with Python** | Cisco Networking Academy
- **IBM Data Analytics** | IBM with MCIT Egypt
- **Introducing Generative AI with AWS** | Udacity
- **Machine Learning Foundations** | AWS Educate
- **AI & Machine Learning Foundations** | SprintUp by Sprints

Languages

- **Arabic** Native Speaker
- **English** Professional Working Proficiency